

CIBAO INTL, Dominican Republic

WMO: 784600

Lat: 19.406N

Lon: 70.605W

Elev: 565

StdP: 14.40

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.3	64.0	59.2	76.6	74.5	60.9	81.6	73.5	18.9	82.4	17.6	82.1	0.7	290	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.0	93.4	76.8	92.5	76.9	91.5	77.0	80.4	87.5	79.7	86.8	79.1	86.3	13.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	153.5	82.6	78.0	148.5	81.9	77.3	144.8	81.4	44.5	88.0	43.7	86.8	43.0	85.7	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.8	16.1	DB	57.6	95.8	2.7	2.1	55.6	97.2	54.0	98.5	52.5	99.6	50.5	101.1	(4)
(5)			WB	57.1	84.2	2.6	2.9	55.2	86.2	53.6	87.9	52.2	89.5	50.2	91.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.9	74.4	75.3	76.4	78.4	79.9	81.8	82.1	82.1	81.8	80.7	77.8	75.5	(6)	
	DBStd	3.60	2.33	2.51	3.21	2.45	2.08	1.79	1.76	2.04	1.74	2.07	2.41	2.43	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	10541	756	709	818	853	928	955	994	997	954	951	835	791	(10)	
	CDD65	5066	291	289	353	403	463	505	529	532	504	486	385	326	(11)	
	CDH74	41188	1706	1896	2567	3077	3703	4718	5104	5090	4593	4086	2687	1961	(12)	
(13)	CDH80	15997	428	571	894	1159	1425	2040	2193	2210	2015	1659	884	519	(13)	
(14)	Wind	WSAvg	7.1	5.7	6.6	6.9	7.7	8.0	9.1	8.5	6.7	6.0	5.0	5.5	(14)	
(15) Precipitation	PrecAvg	45.3	2.5	2.6	2.6	4.1	5.8	2.9	4.1	4.1	4.1	4.8	5.3	4.9	(15)	
	PrecMax	61.8	5.3	7.0	6.9	9.0	15.3	6.9	9.8	9.8	8.0	9.1	14.7	12.9	(16)	
	PrecMin	33.0	0.3	0.2	0.6	0.2	0.9	0.9	0.9	1.5	0.6	0.6	0.6	0.7	(17)	
	PrecStd	8.5	1.3	2.0	1.6	2.5	3.9	1.7	2.3	2.1	2.2	2.4	3.3	3.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.5	89.4	91.6	91.7	92.9	93.5	93.6	95.3	94.8	93.4	90.0	88.2	(19)	
		MCWB	72.8	73.8	74.0	74.9	74.8	77.1	76.8	76.3	76.6	76.7	76.2	74.5	(20)	
	2%	DB	85.8	87.4	89.5	89.8	91.0	92.6	92.8	93.3	92.9	91.3	88.4	86.3	(21)	
		MCWB	72.3	72.8	73.1	74.1	75.8	77.0	77.0	77.0	77.3	77.2	75.9	74.4	(22)	
	5%	DB	84.3	85.9	87.7	88.2	89.4	91.4	91.4	91.6	91.4	89.8	87.5	84.6	(23)	
		MCWB	72.3	72.4	72.7	73.9	75.8	77.1	77.2	77.4	77.5	77.0	75.8	74.0	(24)	
	10%	DB	82.6	84.1	85.7	86.4	87.6	89.7	89.8	89.9	89.7	88.1	85.7	83.0	(25)	
		MCWB	72.4	71.9	72.5	73.8	75.8	77.0	77.0	77.4	77.4	77.0	75.7	73.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.5	76.4	77.3	78.8	79.9	81.0	81.0	81.7	80.9	80.4	79.0	78.0	(27)	
		MCDB	81.5	83.9	85.3	86.2	86.5	87.9	87.9	87.6	88.6	86.8	84.7	83.1	(28)	
	2%	WB	75.2	75.0	75.7	77.2	78.8	79.8	79.8	80.0	79.7	79.2	78.0	76.7	(29)	
		MCDB	80.3	81.8	83.1	84.7	85.1	87.1	86.9	87.5	87.3	86.0	83.9	81.9	(30)	
	5%	WB	74.3	74.1	74.7	76.0	77.8	79.0	79.0	79.1	79.0	78.7	77.2	75.7	(31)	
		MCDB	79.6	80.9	81.9	83.3	84.2	86.4	86.2	86.6	86.6	85.5	83.3	81.1	(32)	
	10%	WB	73.6	73.3	73.8	75.1	77.0	78.3	78.3	78.5	78.3	78.0	76.5	74.9	(33)	
		MCDB	79.0	80.0	80.8	82.2	83.4	85.5	85.4	85.8	85.8	84.7	82.4	80.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	17.8	18.2	17.8	16.6	17.3	17.0	17.0	18.1	16.8	15.9	16.5	(35)	
		MCDBR	19.3	19.9	20.5	19.3	18.3	18.6	18.5	18.7	19.6	18.4	18.2	18.3	(36)	
	5% WB	MCWBR	8.8	7.5	6.7	6.1	5.8	6.1	5.9	5.9	6.8	6.6	7.2	8.4	(37)	
		MCWBR	16.9	17.8	18.4	17.4	16.1	17.1	16.9	17.0	18.1	16.8	15.9	16.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.363	0.377	0.391	0.419	0.445	0.508	0.492	0.469	0.427	0.391	0.383	0.383	(40)	
		taud	2.544	2.493	2.450	2.400	2.348	2.119	2.171	2.264	2.412	2.521	2.532	2.501	(41)	
	Ebn at Noon		291	292	291	282	271	252	257	265	276	285	282	280	(42)	
		Edh at Noon	31	34	36	39	41	50	48	44	37	33	31	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1345	1572	1757	1832	1796	1909	1935	1880	1787	1564	1328	1276	(44)	
(45)		RadStd	94	90	123	138	117	104	100	77	111	116	114	72	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.49	N/A	N/A	N/A	N/A

Nomenclature: See separate page